

EXPLORATORY DATA ANALYSIS REPORT

Ford GoBike 2018 Trip Data

Twelve months of bike-share trip data · San Francisco Bay Area

1,863,721 raw trip records · 1,740,142 records after cleaning
Prepared for internal review

Project Summary

This report analyzes twelve months of Ford GoBike trip data for 2018, combining individual monthly files into a single dataset for data wrangling, exploratory analysis, and visualization. Ford GoBike is a bike-share program operating in the San Francisco Bay Area, and each trip record captures duration, start and end stations, timestamps, and rider demographics (age, gender, user type). The goal of this analysis is to uncover usage patterns that can inform fleet operations, pricing, and marketing decisions.

Business Objective

The primary objective is to help Ford GoBike make data-driven decisions that improve operational efficiency, support revenue growth, and enhance the rider experience. Specifically, the analysis aims to:

- **Optimize fleet operations** — identify peak demand windows and high-traffic stations so bikes can be rebalanced proactively.
- **Support revenue growth** — understand the Subscriber vs. Customer mix and what drives conversion.
- **Improve rider experience** — understand how trip duration and usage patterns differ by rider segment, time of day, and season.

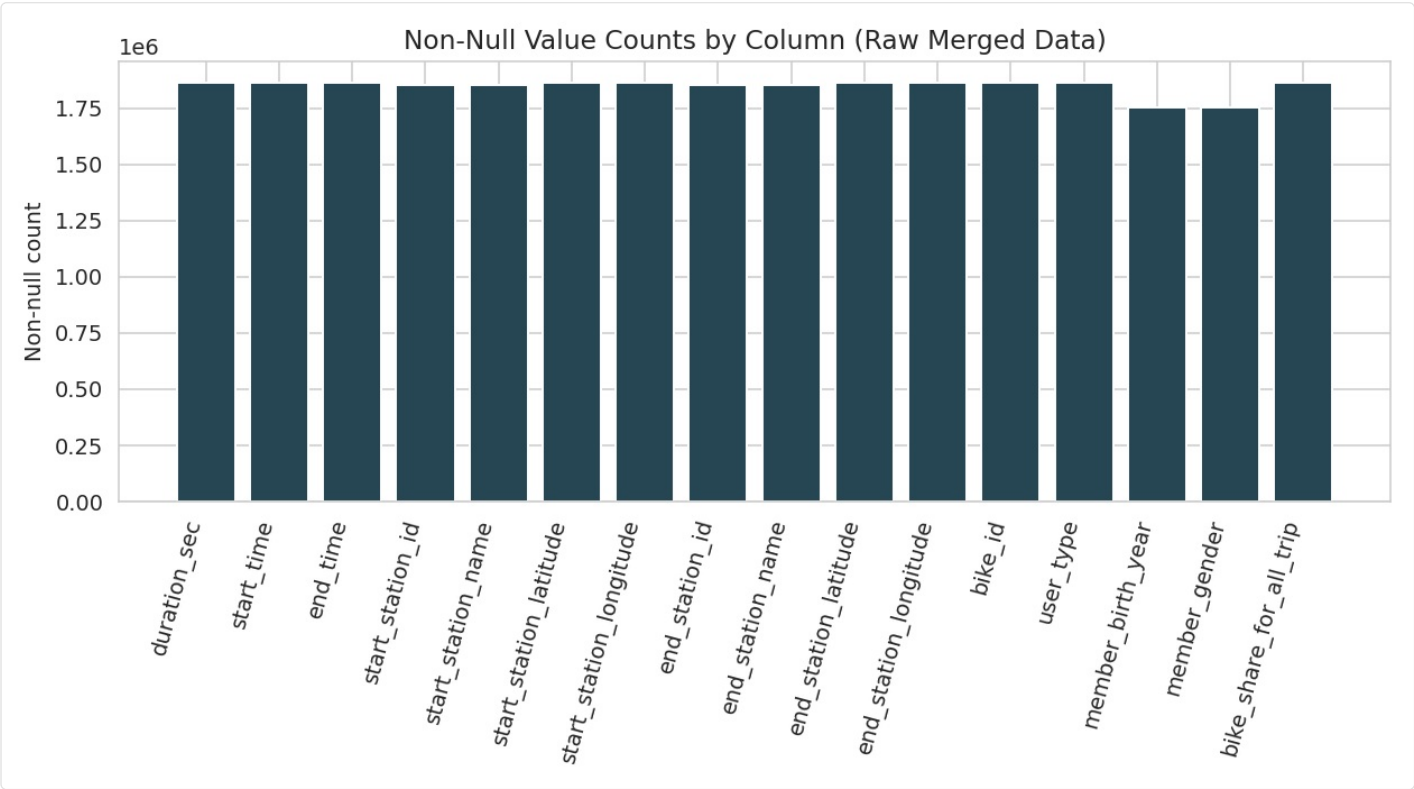
Data Overview & Cleaning

1,863,721 RAW TRIP RECORDS (12 MONTHS)	1,740,142 RECORDS AFTER CLEANING	123,579 ROWS DROPPED	0 DUPLICATE ROWS FOUND
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Key data wrangling steps applied to the raw, merged dataset:

Step	Detail
Type conversion	start_time / end_time parsed to datetime; start_station_id, end_station_id, bike_id cast to string so they aren't treated as numeric; user_type and bike_share_for_all_trip cast to category to reduce memory footprint.
Missing station data	Rows with null start/end station id or name were dropped (these can't be meaningfully analyzed for station-level insights).
Missing gender	Null values in member_gender filled with "Unknown" rather than dropped, to preserve trip volume.
Birth year outliers	Minimum birth year in the raw data was 1881 (implying a 137-year-old rider). 1,413 rows with birth year < 1930 were identified as implausible and dropped. Max birth year of 2000 aligns with an 18-year minimum riding age.

Non-Null Value Counts (Before Cleaning)

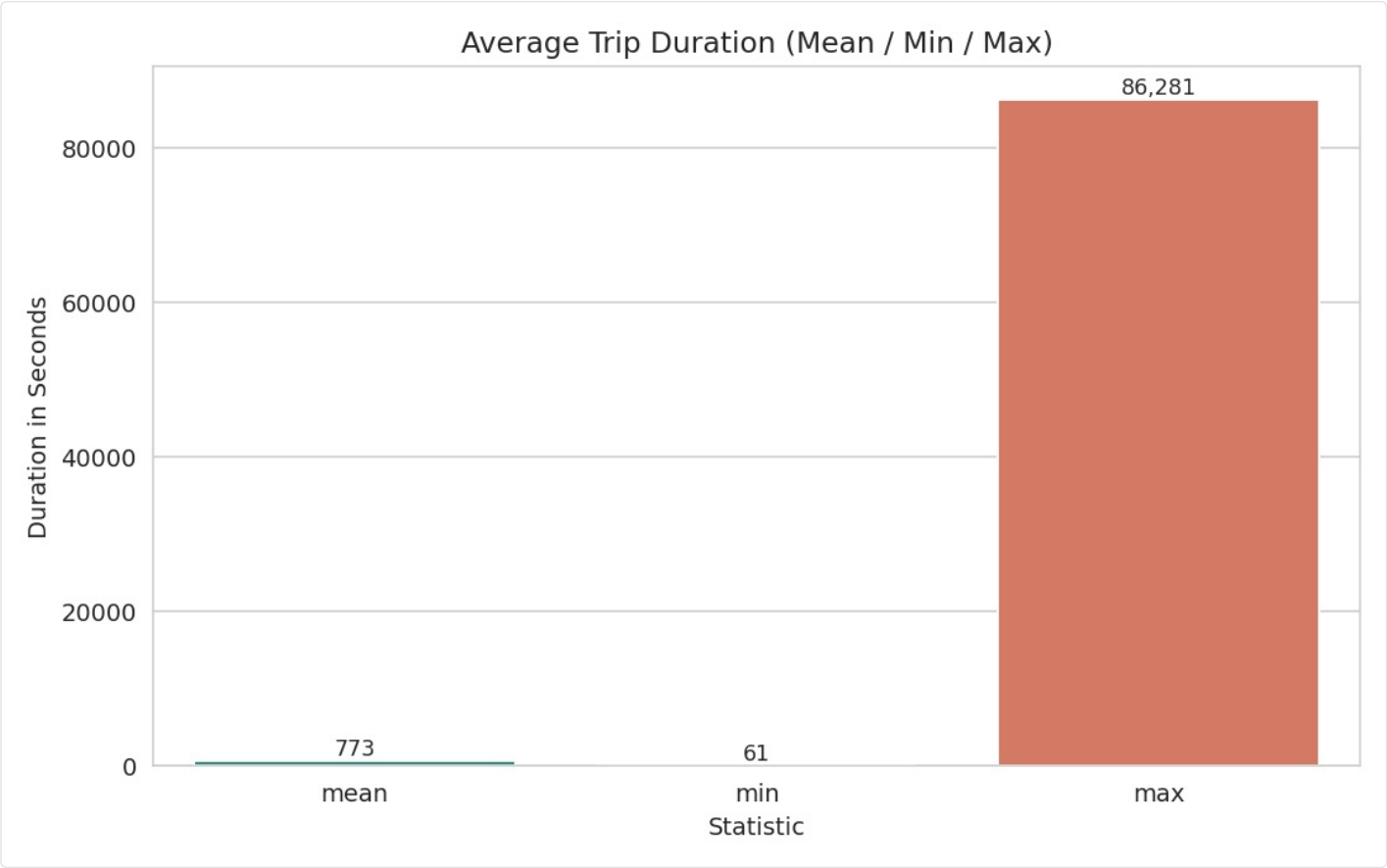


Station id/name and member demographic fields (birth year, gender) carry most of the missingness in the raw data — consistent with Customers (non-subscribers) being less likely to have demographic info on file.

1. Univariate Analysis

Examining individual variables in isolation to understand their distribution before looking at relationships.

1.1 Average Trip Duration

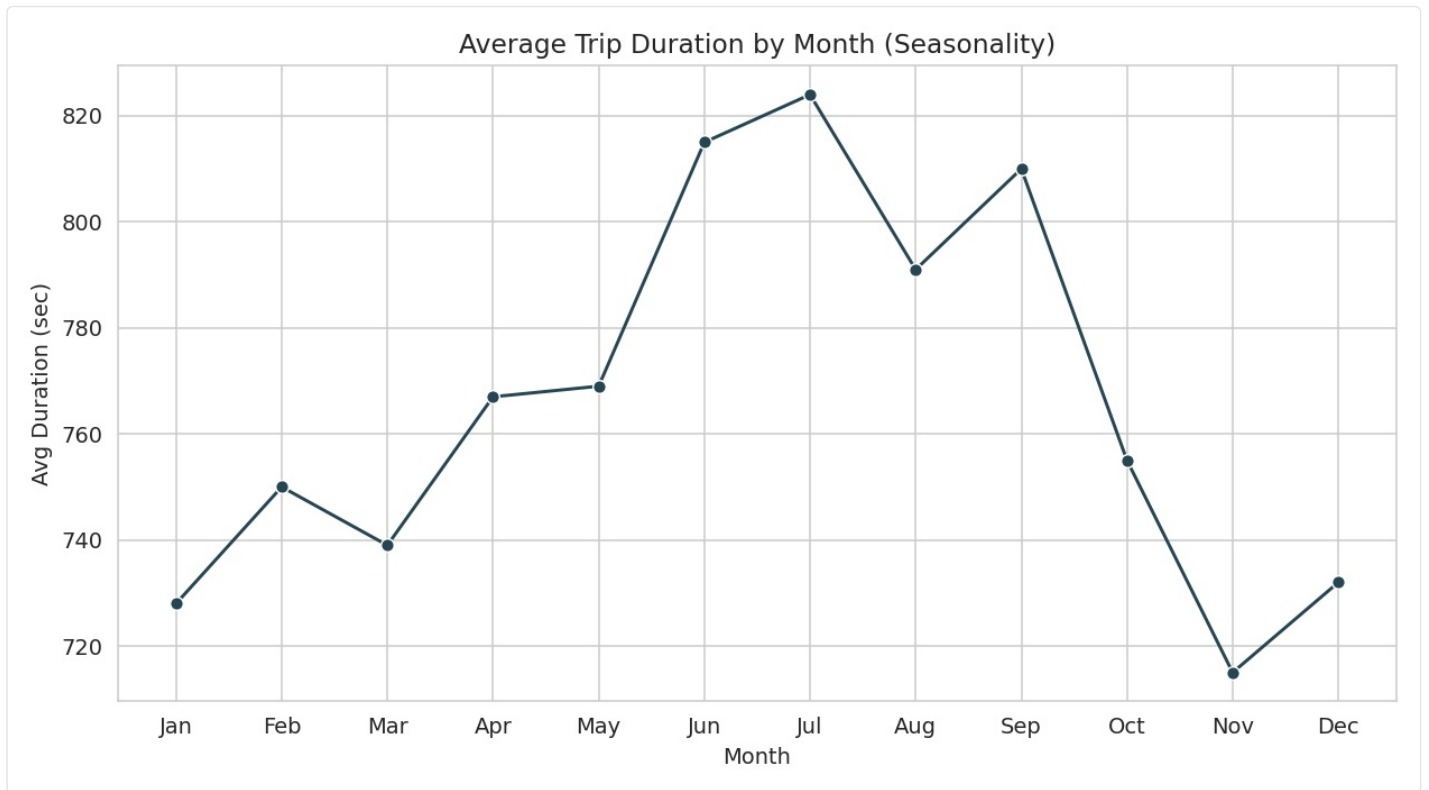


Why this chart: A bar chart clearly compares the mean trip duration against the minimum and maximum, making the spread immediately visible.

Insight: Average trip duration is 773 seconds (~12.9 minutes), while the maximum trip duration is dramatically higher — a strong signal of outliers in the data.

Business impact: Knowing the typical (short) trip length helps size bike availability for quick trips, while the outlier gap flags a need to look separately at long-duration rides (possibly lost, stolen, or leisure-use bikes).

1.2 Trip Duration by Month (Seasonality)

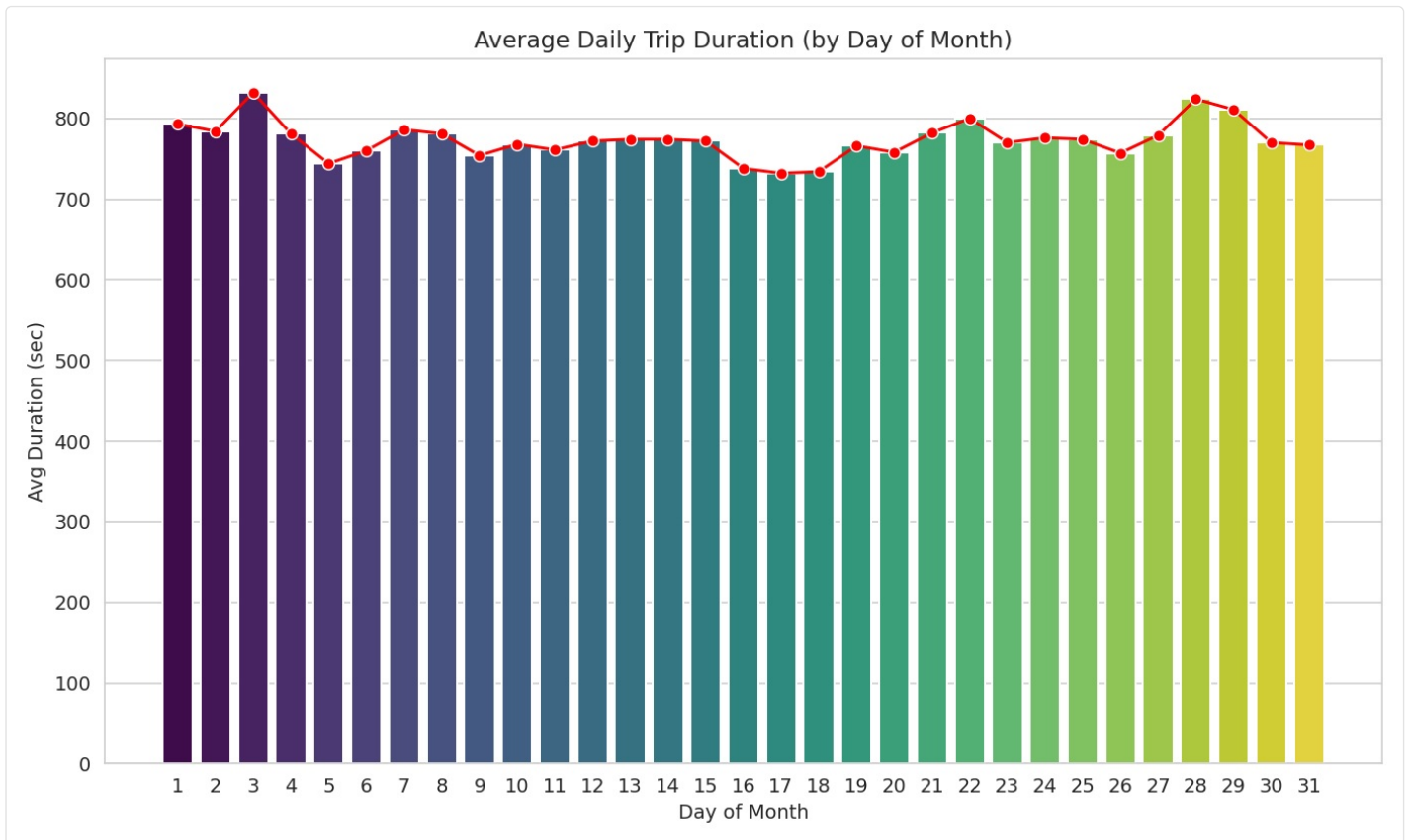


Why this chart: Month is a sequential variable, so a line chart is well suited to showing how average trip duration trends across the year.

Insight: Average trip duration peaks in Jul at 824 seconds, higher than the rest of the year — consistent with warmer-weather, more leisurely summer riding.

Business impact: Seasonal peaks support planning for higher summer fleet demand and timing seasonal promotions or maintenance windows around lower-demand months.

1.3 Daily Trip Duration (by Day of Month)

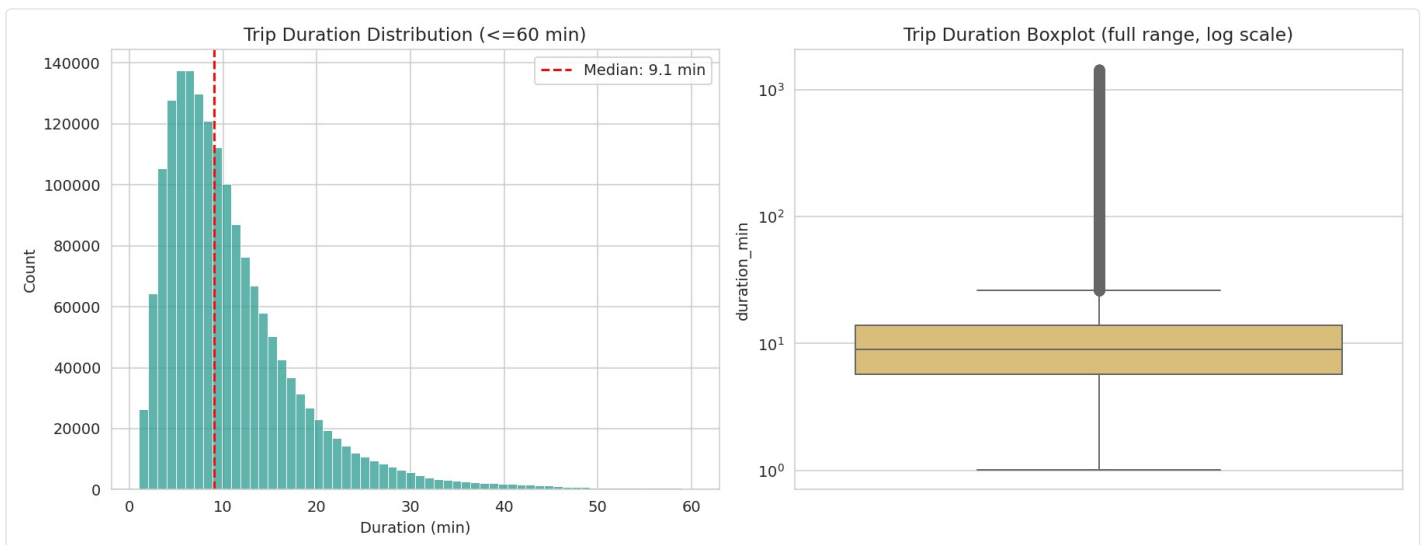


Why this chart: A combined line and bar view over day-of-month shows fluctuation patterns and makes peaks and troughs easy to spot.

Insight: Average duration fluctuates moderately across the days of the month without a strong monotonic trend, suggesting day-of-month itself is a weak driver of trip length compared to day-of-week or season.

Business impact: Because day-of-month shows little signal, operational planning should prioritize day-of-week and seasonal patterns over calendar-day scheduling.

1.4 Trip Duration Distribution

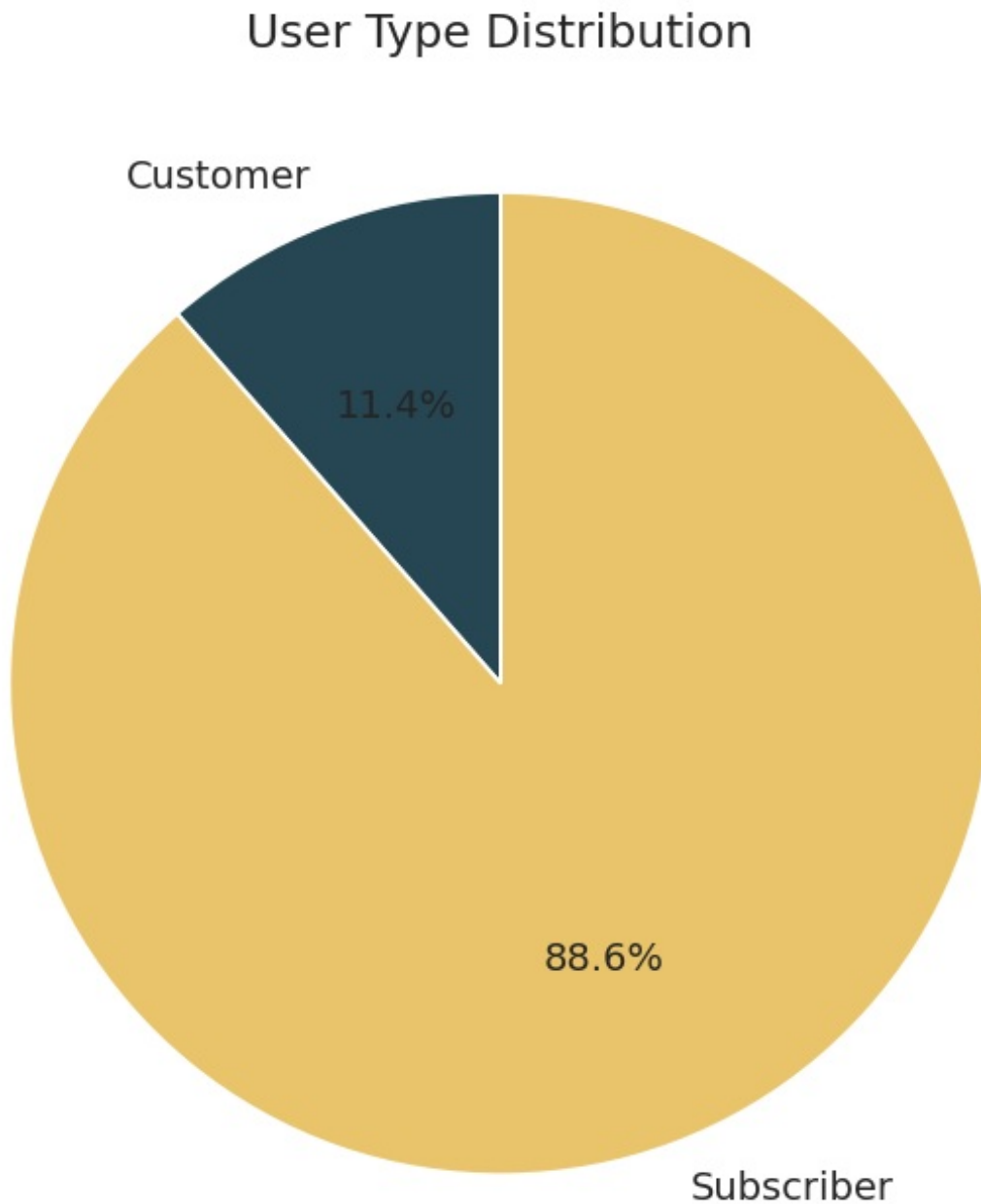


Why this chart: A histogram (capped at 60 minutes) shows the shape of the bulk of trips, while a log-scale boxplot alongside it exposes the extreme long-tail outliers without distorting the view.

Insight: The distribution is heavily right-skewed: the median trip is only 9.1 minutes, with the vast majority of trips well under 20 minutes, confirming GoBike is used mostly for short, functional trips rather than long leisure rides.

Business impact: This supports pricing models built around short trips (e.g. per-minute overage fees beyond a free window) and confirms the fleet should be optimized for high turnover rather than long single-rider holds.

1.5 User Type Distribution

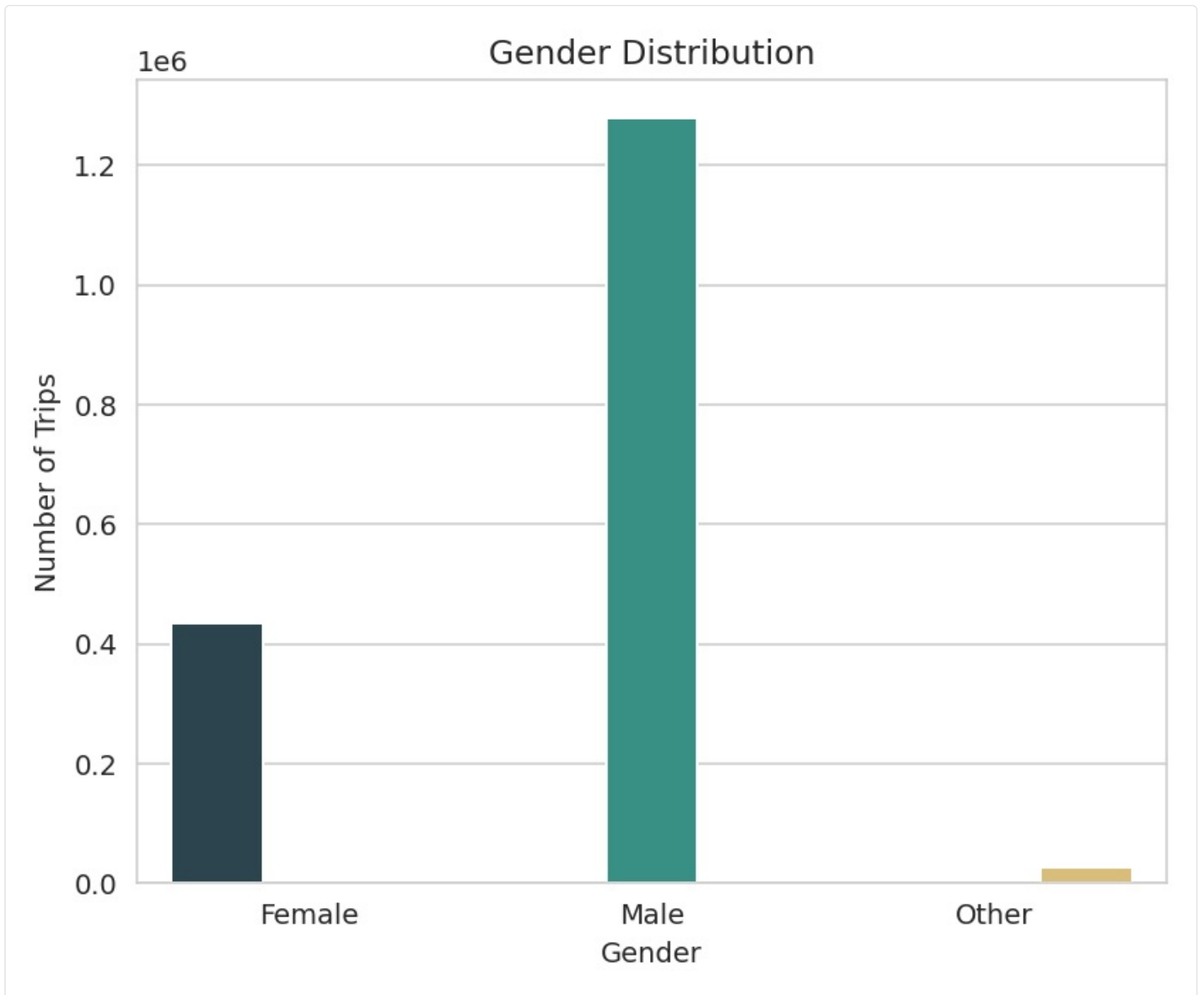


Why this chart: A pie chart suits user_type well since it has only two categories, making the proportion split easy to read at a glance.

Insight: Subscribers make up 88.6% of trips versus 11.4% for Customers — the platform is dominated by recurring, subscribed users.

Business impact: High Subscriber share indicates strong recurring revenue and product-market fit for commuters; growth strategy should focus on converting the smaller Customer segment into Subscribers.

1.6 Gender Distribution

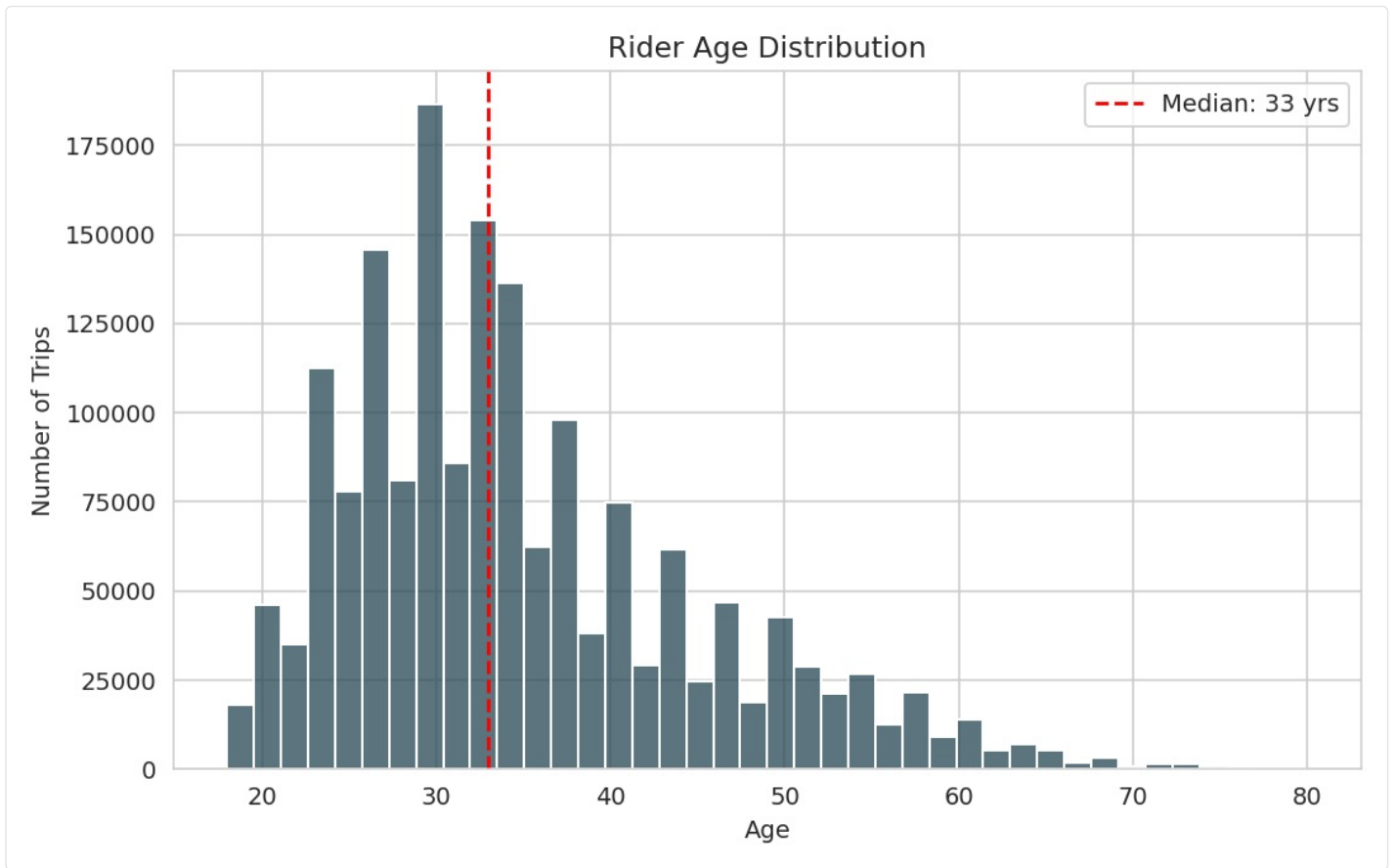


Why this chart: A bar chart suits member_gender since it's categorical, making counts easy to compare across groups.

Insight: Male riders make up the largest share of trips, followed by Female riders, with Other/Unknown a smaller minority.

Business impact: The gender skew suggests an opportunity to investigate and address barriers to adoption among underrepresented rider groups through targeted outreach.

1.7 Rider Age Distribution

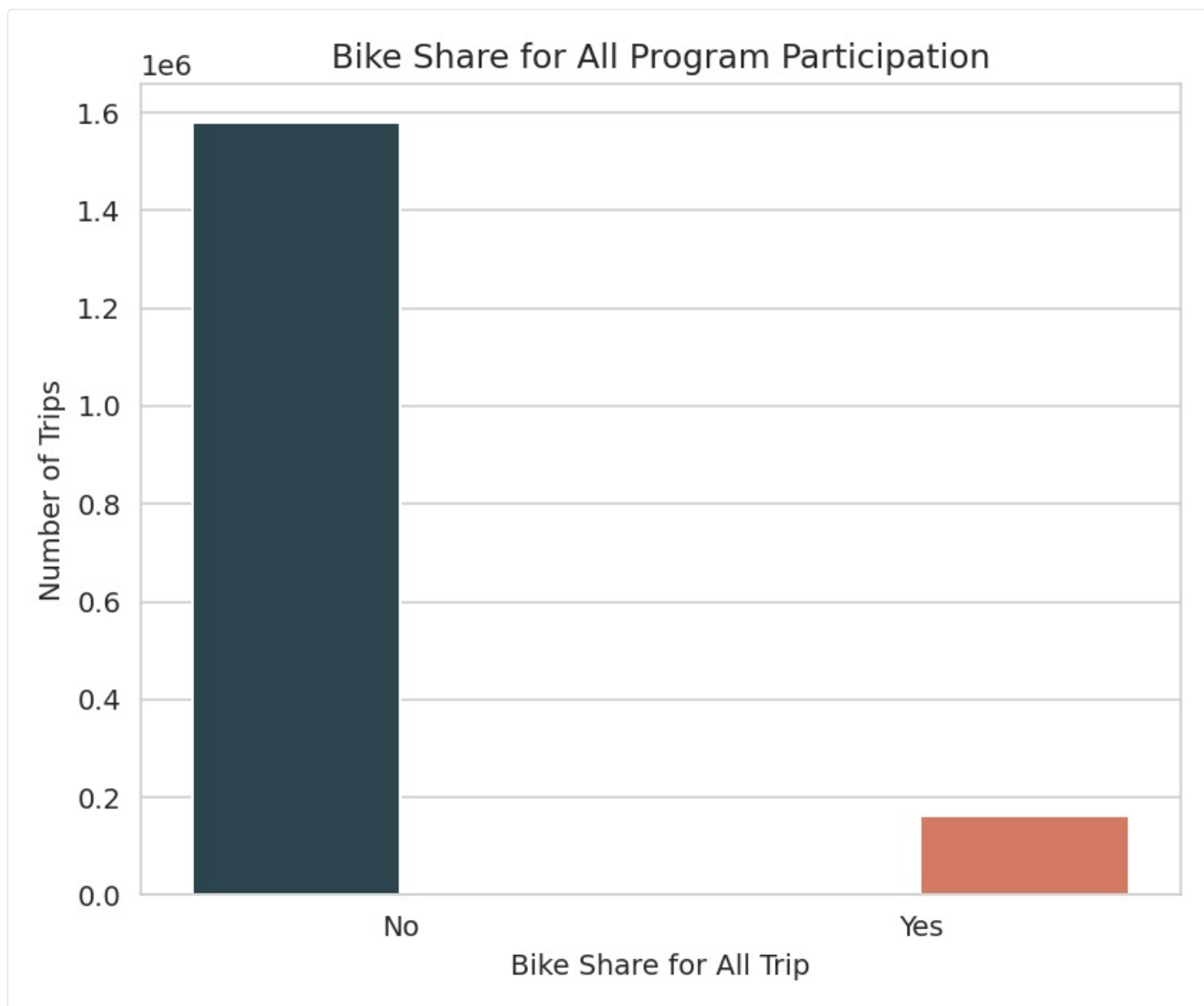


Why this chart: A histogram suits age since it's continuous, showing the full shape of the distribution rather than a single summary statistic.

Insight: Rider age is concentrated around 30-35 years old, with a median age of 33, tapering off at older ages.

Business impact: A concentrated, working-age rider base supports commuter-focused marketing and partnerships (e.g. employer transit benefits) targeted at that demographic.

1.8 Bike Share for All Program Participation



Why this chart: A count plot suits this binary Yes/No field, making participation counts easy to compare directly.

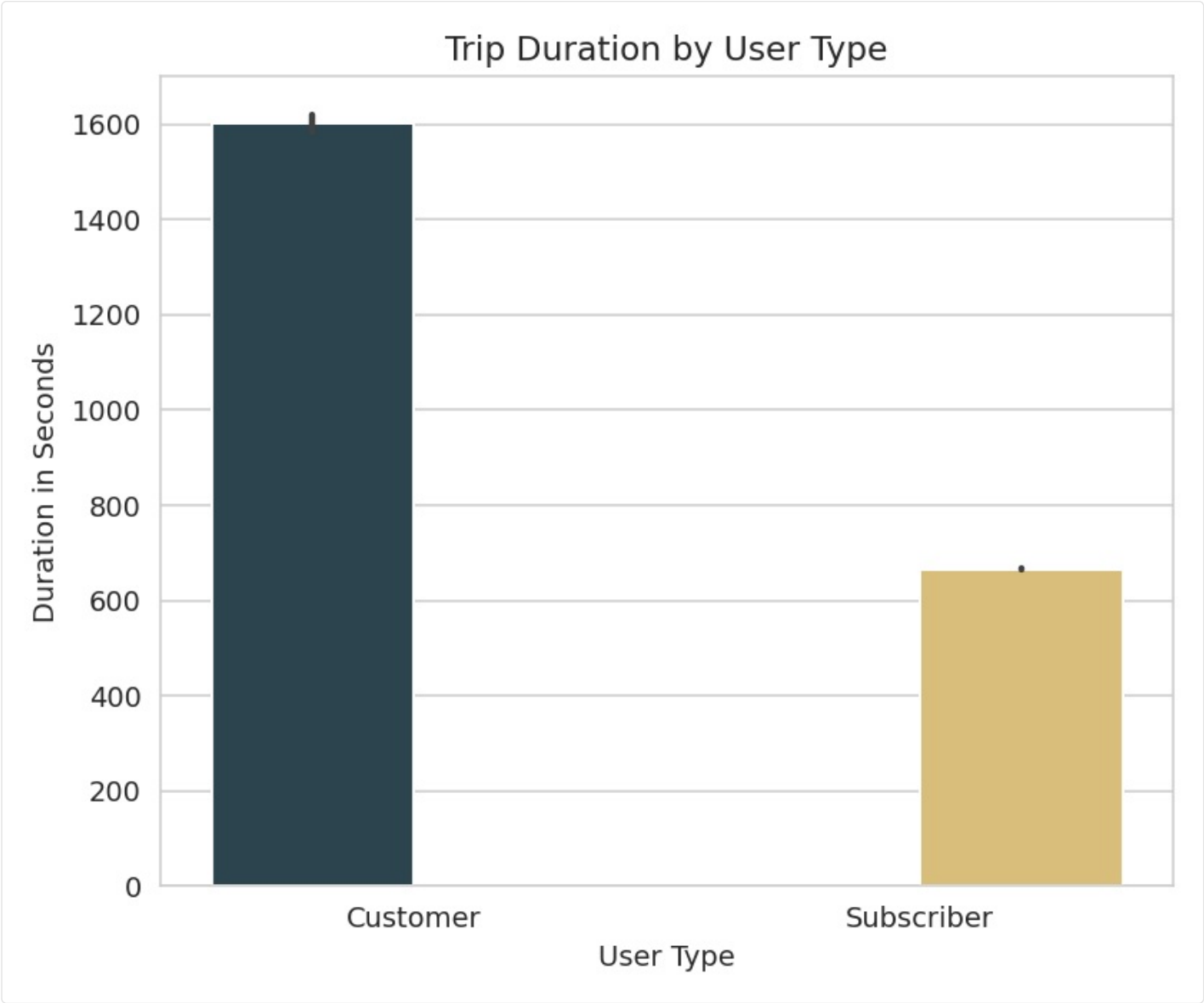
Insight: The large majority of trips are not part of the Bike Share for All (subsidized access) program, with only a small share coming from enrolled riders.

Business impact: Low enrollment suggests an opportunity to grow ridership among income-qualified riders through greater awareness of the subsidized program.

2. Bivariate Analysis

Examining relationships between pairs of variables, primarily how trip duration and volume vary by user segment and time.

2.1 Trip Duration by User Type

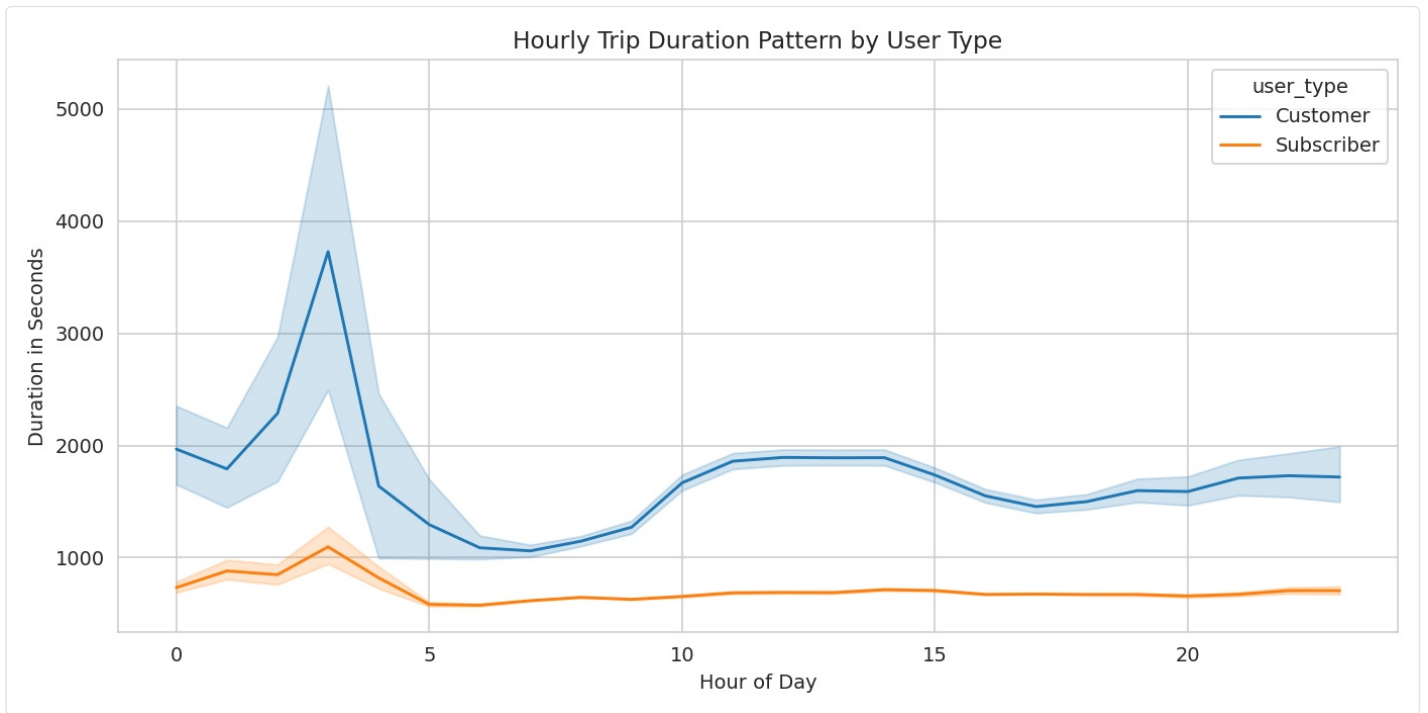


Why this chart: A bar chart directly compares average duration across the two user_type categories.

Insight: Customers have a noticeably higher average trip duration than Subscribers, suggesting Subscribers use bikes for quick commutes while Customers ride more for leisure.

Business impact: Confirms that Subscriber and Customer segments should be treated differently in pricing and marketing — commute-focused messaging for Subscribers, leisure-focused for Customers.

2.2 Hourly Trip Duration Pattern by User Type

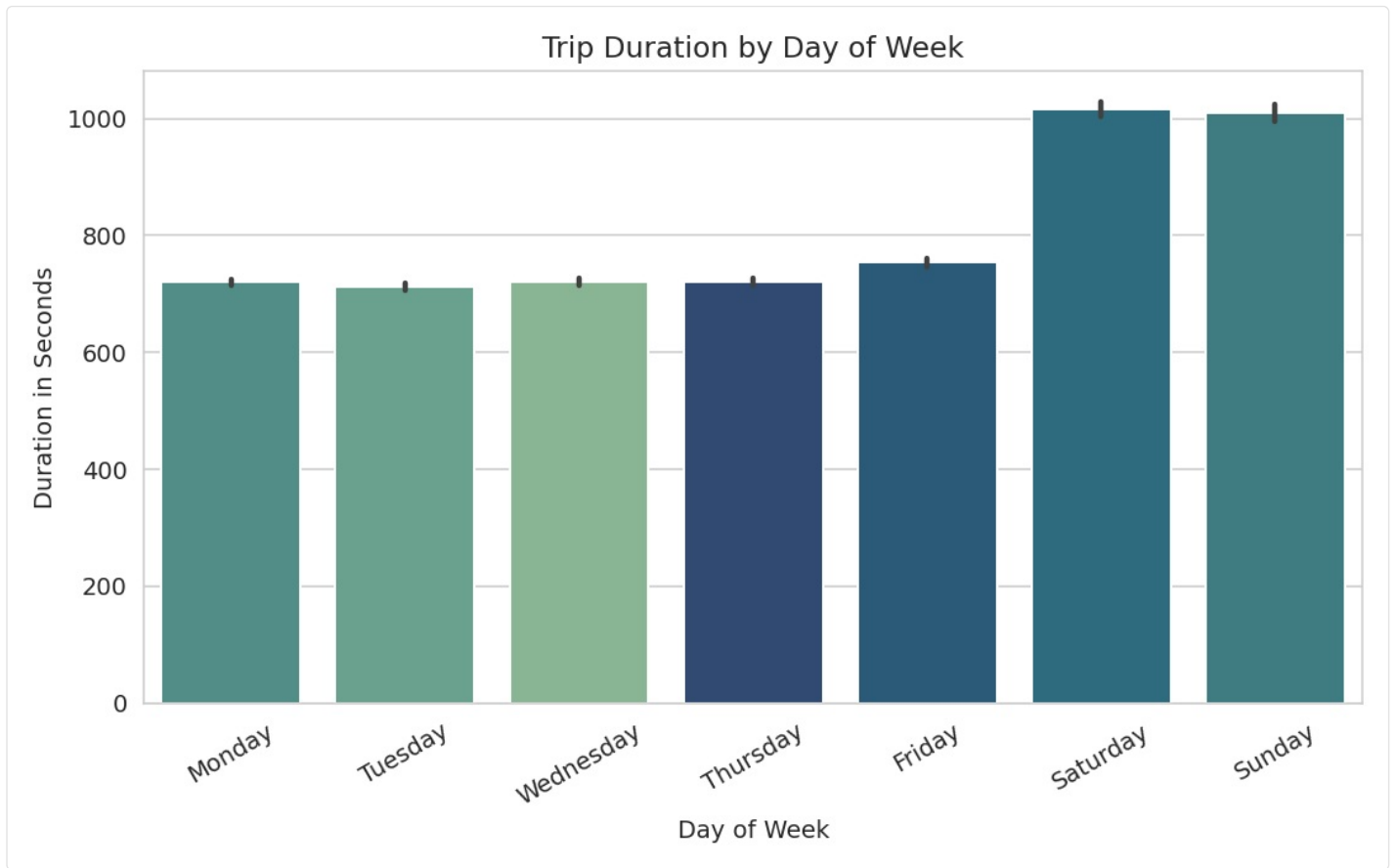


Why this chart: A line chart split by `hue=user_type` shows how the duration trend across hours of the day differs by segment.

Insight: Subscribers show short, steady durations across nearly all hours, while Customers show longer and more variable durations, especially outside typical commute hours.

Business impact: Reinforces that Subscriber demand is commute-driven and predictable, while Customer demand needs different fleet planning for off-peak and weekend windows.

2.3 Trip Duration by Day of Week

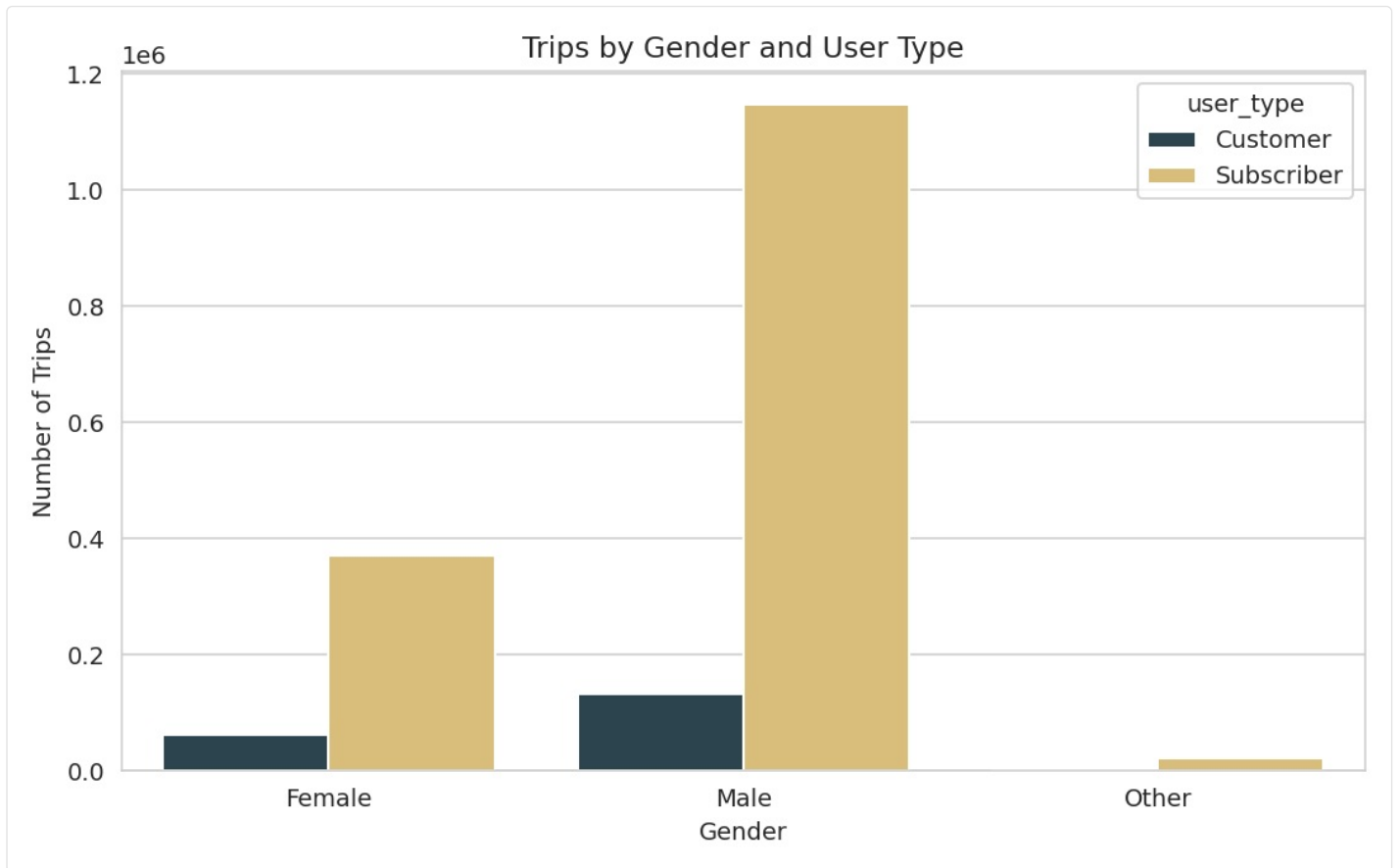


Why this chart: A bar chart compares average duration across each day of the week directly.

Insight: Trip duration is higher on weekends (Saturday, Sunday) compared to weekdays, suggesting more leisure riding on weekends.

Business impact: Supports running weekend-specific promotions targeted at leisure riders, and adjusting weekday fleet rebalancing around shorter, high-frequency commute trips.

2.4 Trips by Gender and User Type

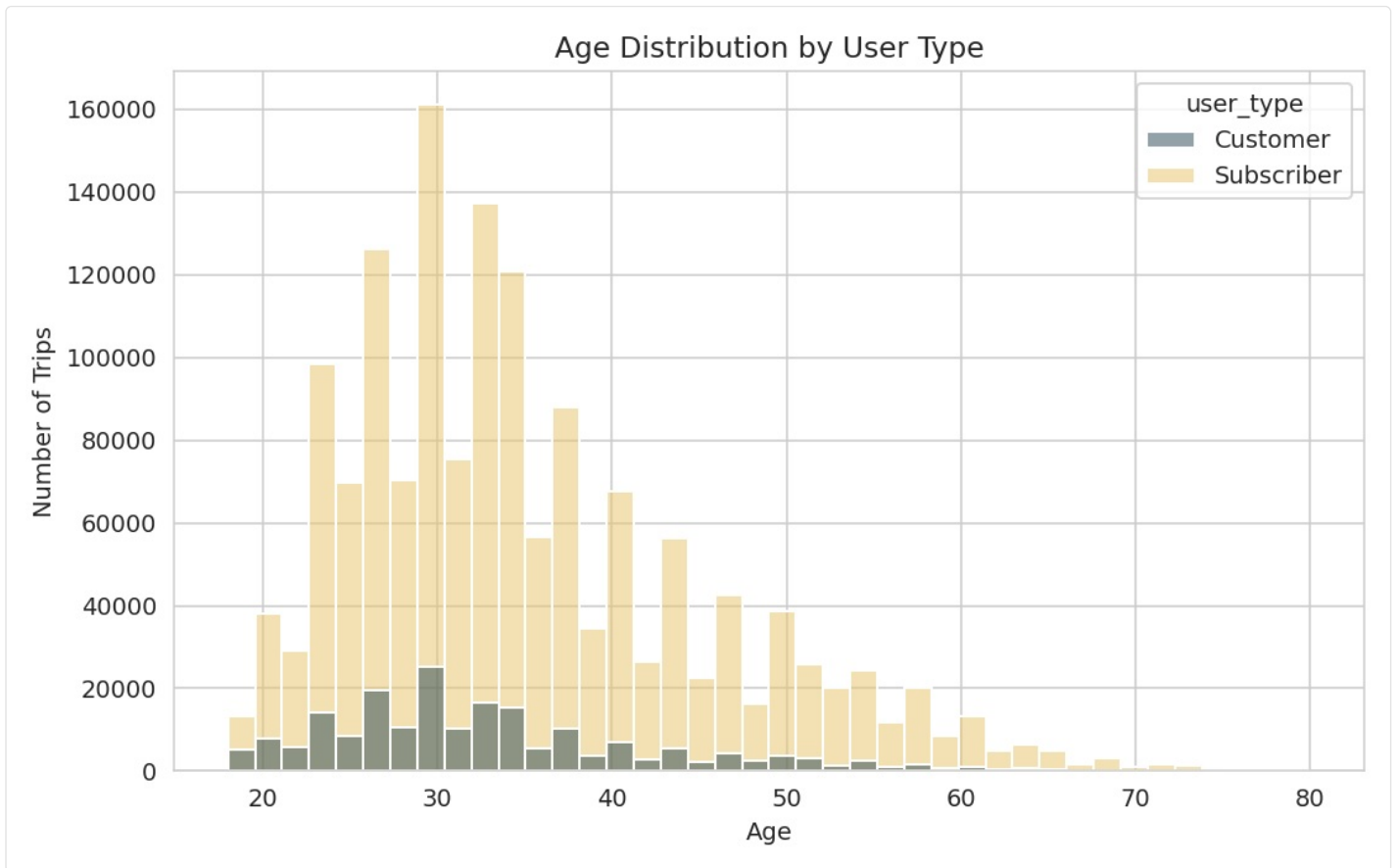


Why this chart: A count plot with `hue=user_type` compares trip counts across gender, split further by user segment.

Insight: Male riders have the highest trip counts in both Subscriber and Customer groups, with Subscribers outnumbering Customers across all gender categories.

Business impact: The consistent Subscriber lead across genders confirms the subscription model is the primary growth lever regardless of demographic segment.

2.5 Age Distribution by User Type

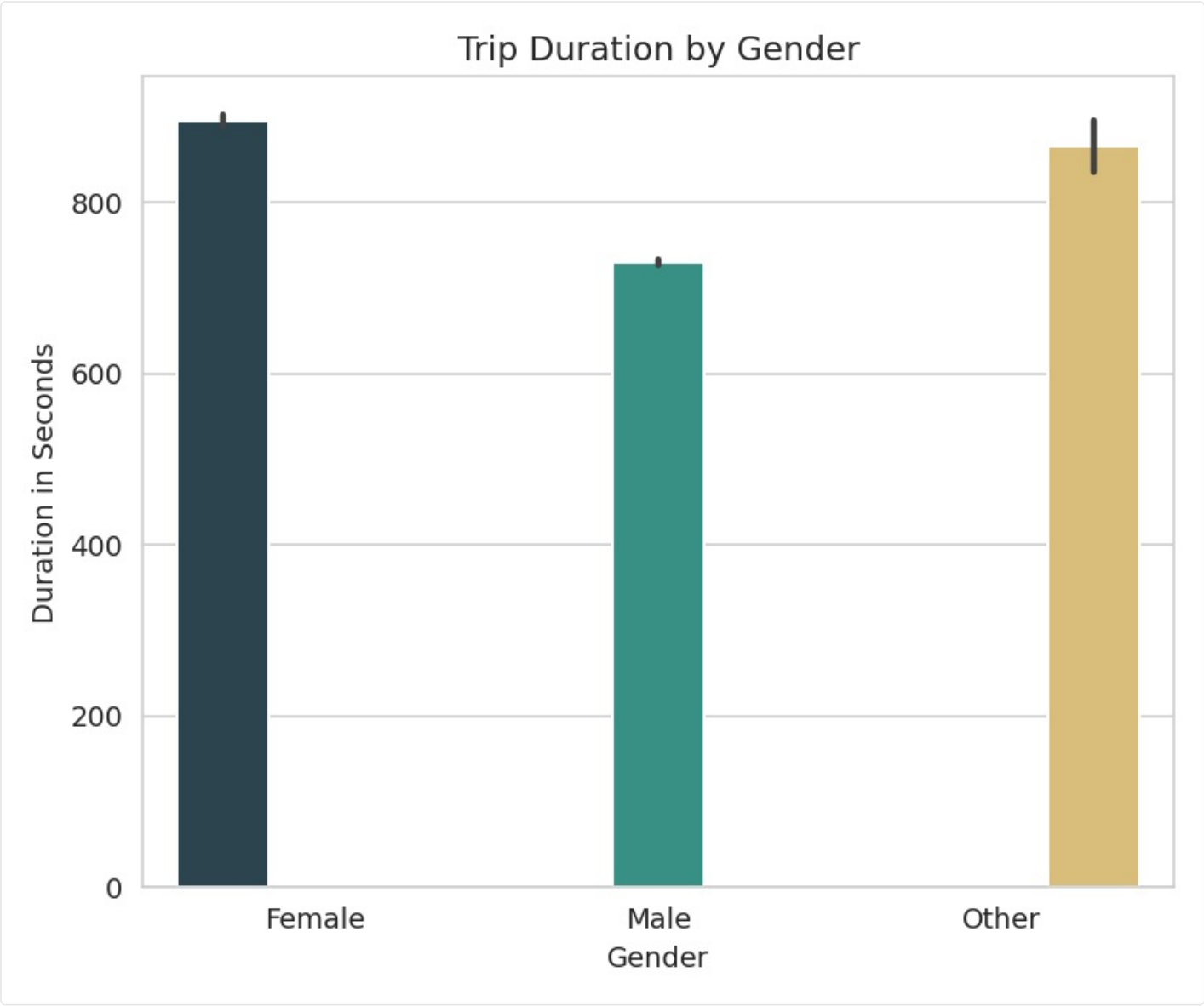


Why this chart: A histogram with `hue=user_type` compares the shape of the age distribution between the two user types directly.

Insight: Subscribers are concentrated in a narrower, younger age range, while Customers show a wider age spread including older riders.

Business impact: The Subscriber core age group is a strong target for continued commuter-focused acquisition, while the broader Customer age range suggests leisure appeal across generations.

2.6 Trip Duration by Gender

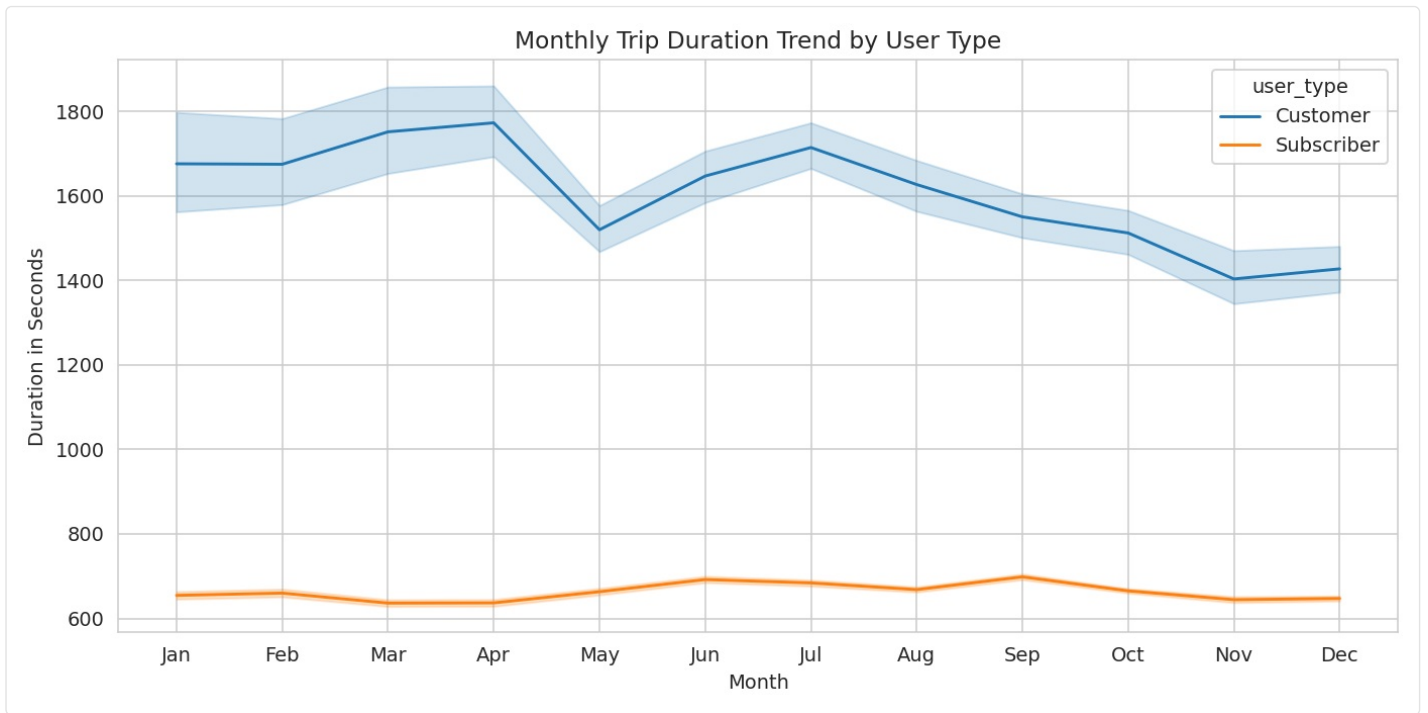


Why this chart: A bar chart compares average duration across gender categories directly.

Insight: Average trip duration is broadly similar across genders, with only minor differences.

Business impact: Since trip length isn't a differentiating factor by gender, pricing and marketing strategy don't need to be gender-segmented on this dimension.

2.7 Monthly Trip Duration Trend by User Type



Why this chart: A line chart tracks the duration trend across months, split by user_type, to see if seasonality affects the two segments differently.

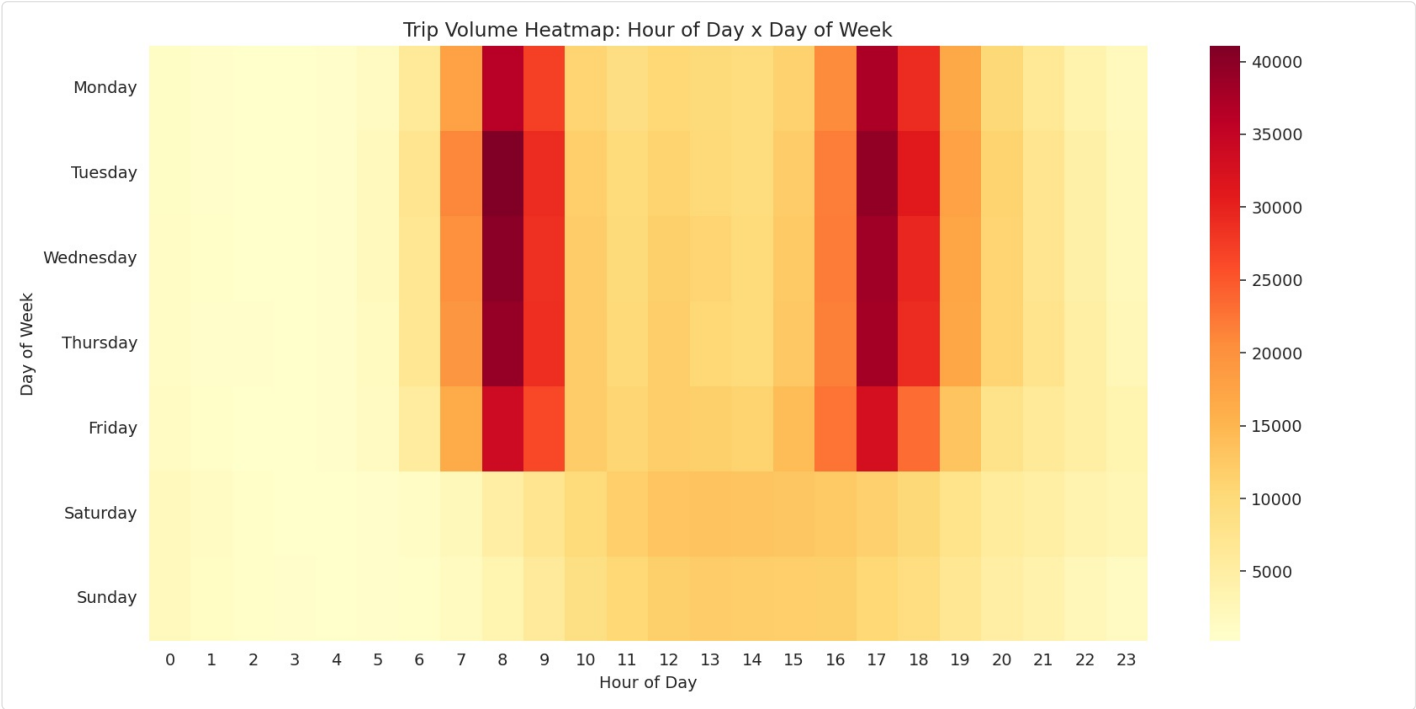
Insight: Customer duration rises more noticeably in summer months, while Subscriber duration stays relatively flat year-round.

Business impact: Supports timing seasonal marketing pushes for Customers around summer, while Subscriber-focused operations can be planned on a flatter, more predictable year-round basis.

3. Multivariate Analysis

Looking at combinations of variables together to uncover deeper structural patterns in demand and rider behavior.

3.1 Trip Volume Heatmap: Hour x Day of Week

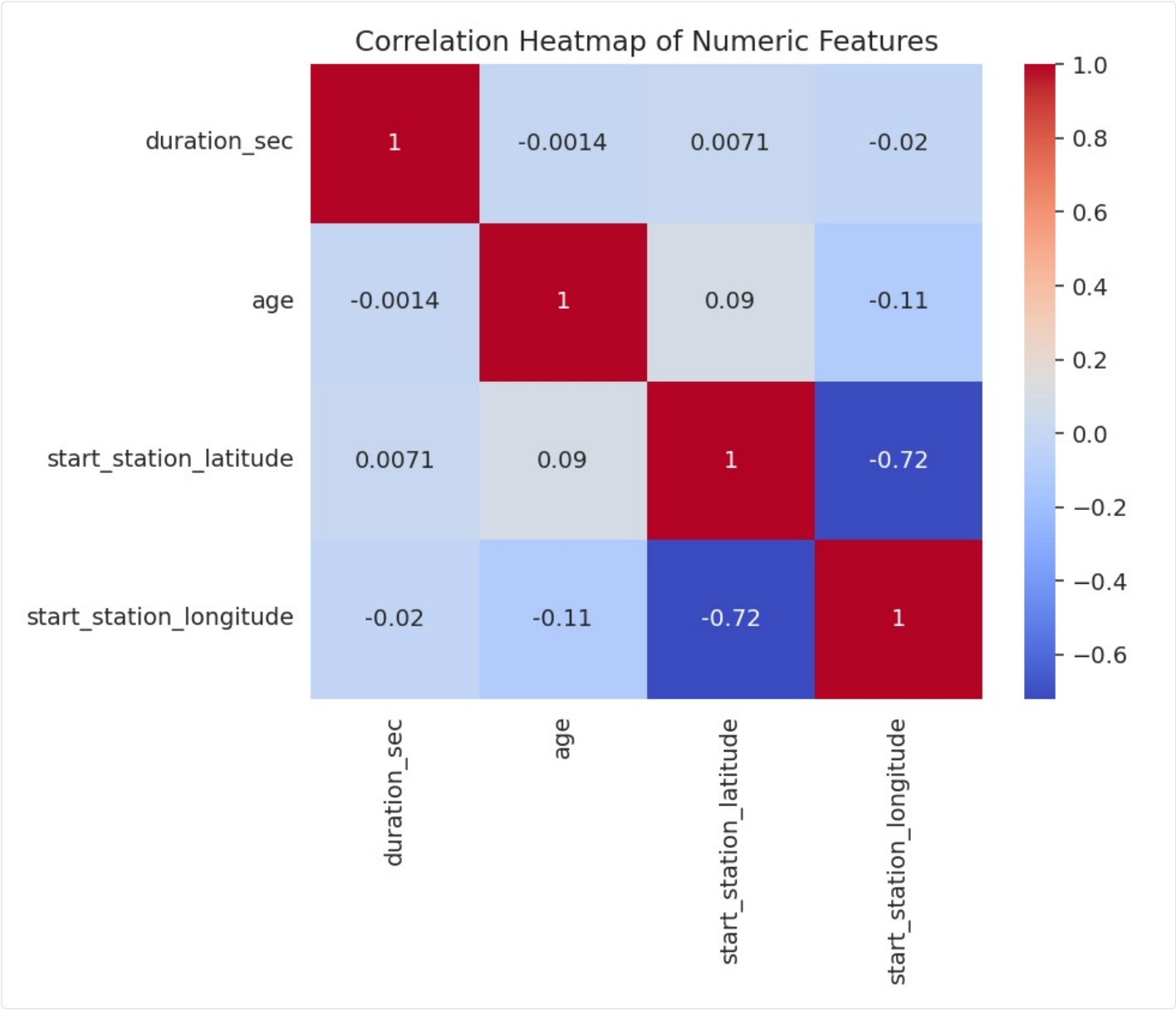


Why this chart: A heatmap shows trip counts across two categorical dimensions (hour and day) at once, with color intensity making peak times easy to spot.

Insight: Trip counts are highest on weekday mornings and evenings (around 8 AM and 5 PM), while weekends show a more spread-out, lower-intensity pattern — a classic commute signature.

Business impact: This is the clearest operational signal in the dataset: fleet rebalancing should be concentrated around weekday morning and evening rush windows at high-traffic stations.

3.2 Correlation Heatmap of Numeric Features

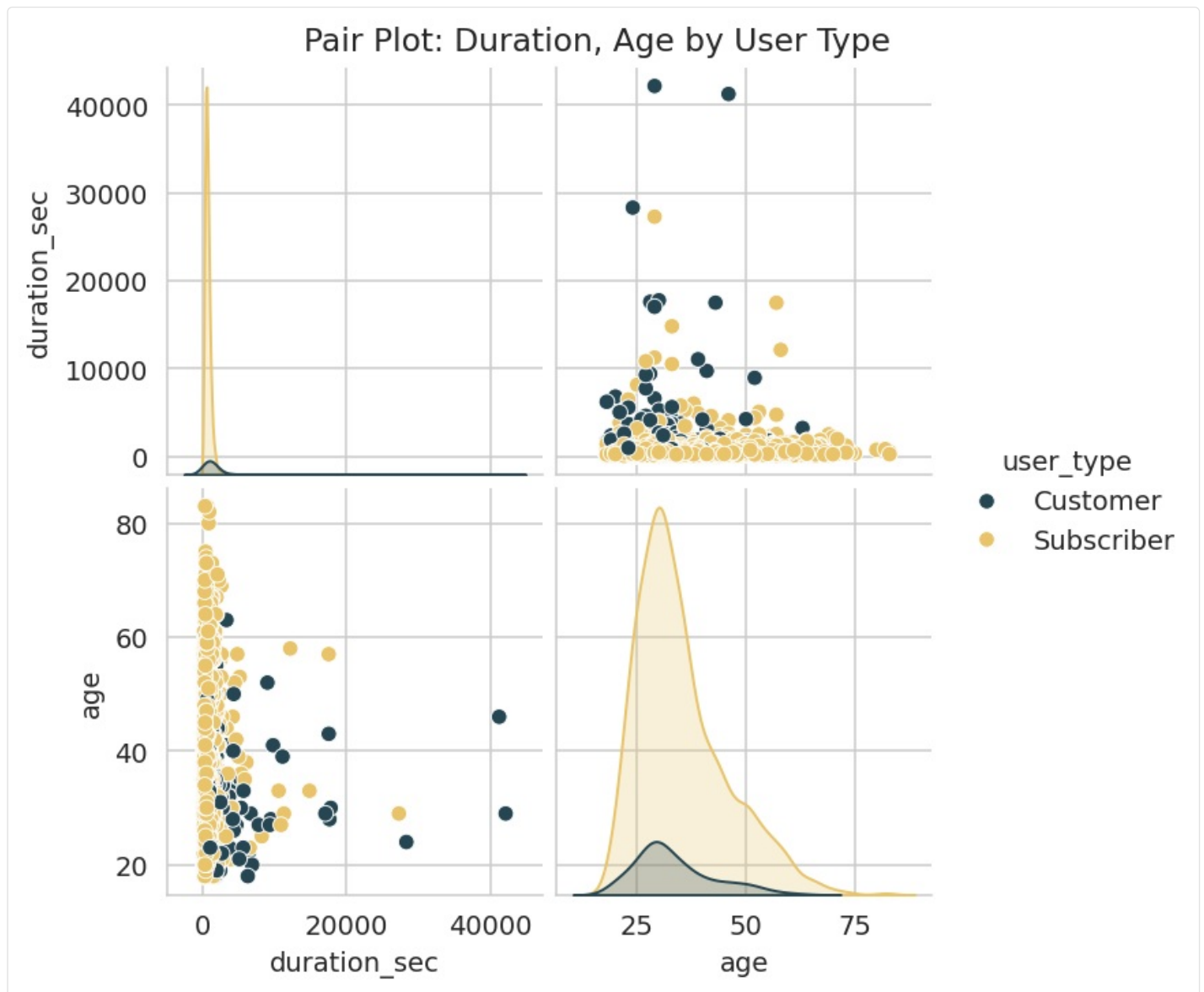


Why this chart: A heatmap with annotations shows correlation strength between all numeric variables at once, in a single view.

Insight: Most numeric variables show weak correlation with each other; duration and age in particular show little to no strong linear relationship.

Business impact: Confirms that trip duration is not simply explained by rider age or station geography alone — behavioral factors like user_type and time-of-day matter more, which should guide future feature engineering.

3.3 Pair Plot: Duration, Age by User Type



Why this chart: A pair plot shows relationships between multiple numeric variables at once, split by user_type, useful for spotting patterns across combinations that a single chart would miss.

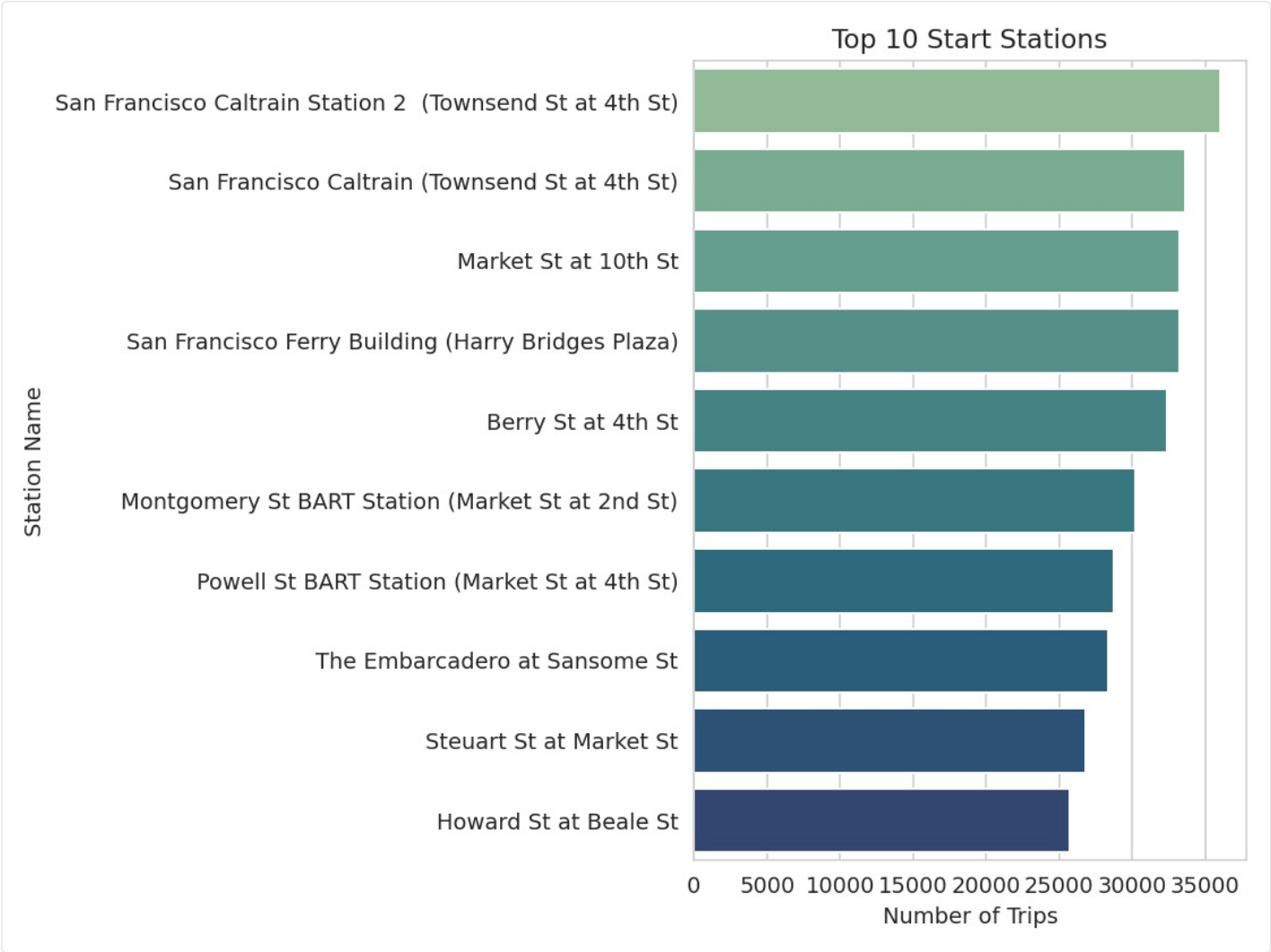
Insight: Duration and age don't show a clear linear relationship, but Subscribers and Customers form somewhat separate clusters, especially on the duration axis.

Business impact: The visible segment separation supports treating Subscriber and Customer as fundamentally distinct rider personas in future modeling and targeting work.

4. Station Analysis

Identifying the busiest locations in the network to guide fleet distribution and station-level operations.

4.1 Top 10 Start Stations

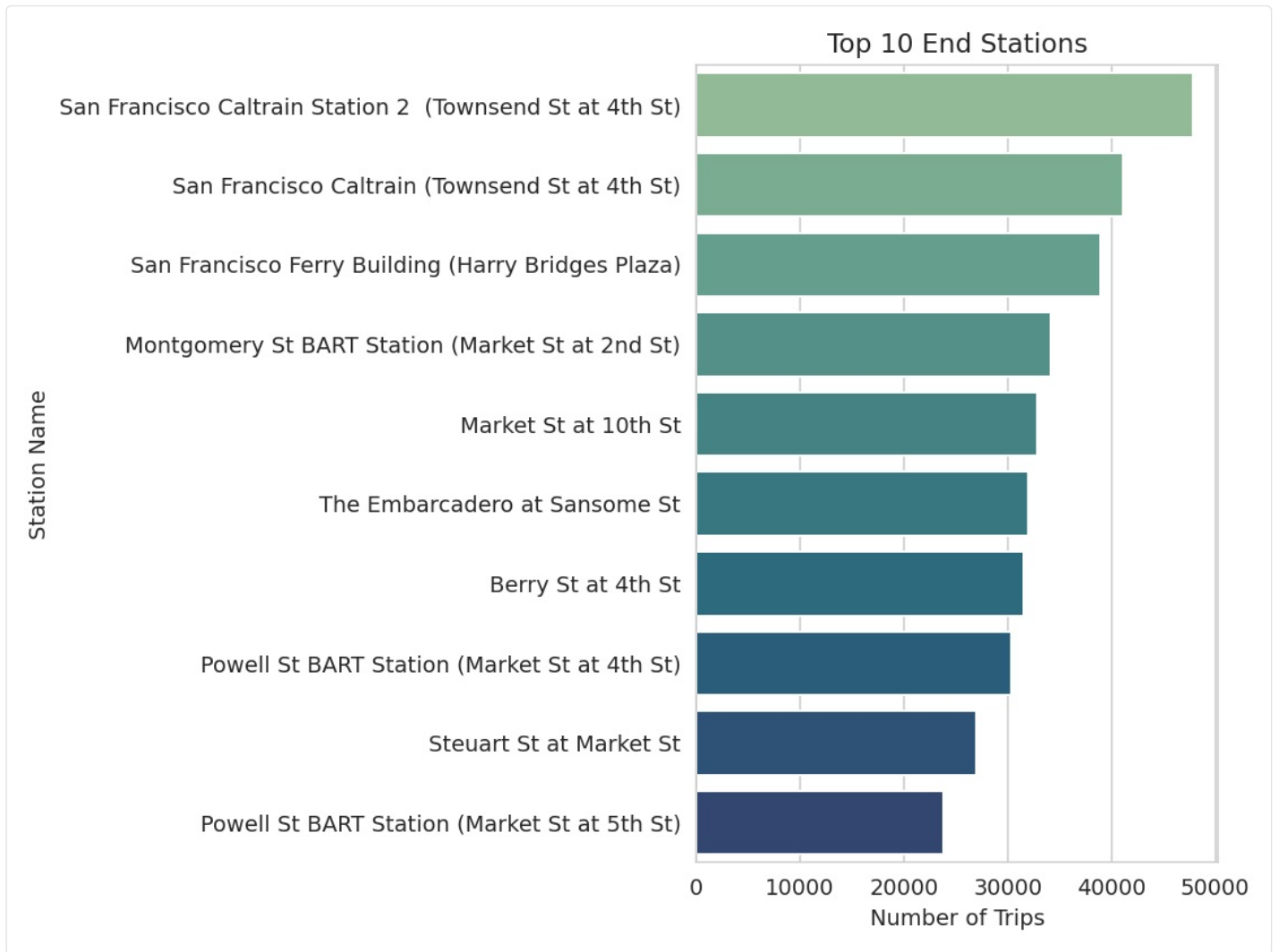


Why this chart: A horizontal bar chart ranks the top 10 start stations by trip count, keeping station names readable alongside their values.

Insight: A small number of stations account for a disproportionately high share of trip starts — led by San Francisco Caltrain Station 2 (Townsend St at 4th St) — likely reflecting high-traffic transit and business hubs.

Business impact: These top stations should be prioritized for proactive bike rebalancing and maintenance to avoid stockouts during peak demand.

4.2 Top 10 End Stations



Why this chart: A horizontal bar chart in the same format as the start-station view enables direct visual comparison between where trips begin and end.

Insight: The top end stations largely overlap with the top start stations (led by San Francisco Caltrain Station 2 (Townsend St at 4th St)), suggesting these are central hubs where trips both begin and conclude.

Business impact: Overlap between top start and end stations confirms these are true network hubs, worth prioritizing for capacity planning, docking infrastructure, and monitoring.

5. Recommendations

- **Rebalance bikes around peak demand windows.** Trip activity spikes sharply on weekday mornings (~8 AM) and evenings (~5 PM) and concentrates at a small number of top stations — prioritize redistributing bikes to these stations ahead of rush hour.
- **Grow the Customer-to-Subscriber pipeline.** Customers ride longer and more on weekends/summer; target them with conversion offers once their usage pattern resembles frequent, sustained riding.
- **Plan fleet capacity for short trips.** The median trip is under 10 minutes — optimize for high bike turnover and availability rather than long single-rider holds.
- **Invest in top-hub stations.** A small set of stations drives a disproportionate share of both starts and ends — prioritize docking capacity and maintenance there.
- **Expand the subsidized access program.** Low participation in Bike Share for All suggests room to grow ridership among income-qualified riders through better awareness campaigns.

Conclusion

Twelve months of raw Ford GoBike data were cleaned and merged into a single, analysis-ready dataset of 1,740,142 trips through careful wrangling — correcting data types, handling missing values, and removing implausible outliers. The analysis found that Subscribers ride frequently for short, predictable commute trips, while Customers ride less often but for longer, more leisure-oriented trips concentrated on weekends and in summer months. Demand is highly concentrated around weekday rush hours and a small set of hub stations, giving clear, actionable direction for fleet rebalancing, targeted marketing, and station-level investment.